

Title: Fatigue Links Sociodemographic Risk to Pain Intensity and Spread in Two Surgical Cohorts

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One Sentence Summary: Fatigue is the strongest and most consistent factor linked to how pain intensifies and spreads before surgery.

Abstract: Why some surgical participants experience pain that extends beyond the original site of injury while others do not remains poorly understood. Both pain intensity and widespread pain contribute to recovery and quality of life, yet their psychosocial correlates are often examined separately. Using data from two large pre-surgical cohorts—participants preparing for knee replacement or thoracic surgery—we examined associations between sociodemographic and psychosocial factors, pain intensity at surgical and non-surgical sites, and widespread chronic pain. Across cohorts and outcomes, fatigue showed the strongest and most consistent associations with pain intensity and widespread pain, independent of other measured factors. Fatigue also occupied a central position in statistical association networks and accounted for substantial shared variance among multiple psychosocial variables, including sleep disturbance, depression, stress, and socioeconomic disadvantage. Pain at non-surgical sites was strongly associated with widespread pain and frequently accounted for observed associations between surgical-site pain and widespread pain. Together, these findings highlight robust patterns of association linking fatigue, pain intensity, and widespread pain in pre-surgical populations.

Main Text:

INTRODUCTION

Pain is a multidimensional experience that extends beyond nociception to include emotional, cognitive, and social processes (1). Two core features of pain are its intensity and its tendency to spread beyond the initial site. Although pain intensity remains a primary focus of clinical care and research, widespread pain is often more difficult to predict, explain, and manage (2, 3). Pain that extends beyond the site of injury or surgery is associated with greater disability, poorer quality of life, and worse recovery outcomes (4, 5). Understanding why pain remains localized in some individuals but becomes widespread in others therefore represents an important clinical and scientific challenge, particularly in surgical populations where tissue injury and recovery trajectories are well defined.

Previous research demonstrates that psychosocial factors play a central role in shaping pain outcomes. Emotional distress, anxiety, depression, and maladaptive coping are consistently associated with higher pain intensity and poorer postoperative recovery (5–7). For example, preoperative anxiety robustly predicts greater postoperative pain intensity, and depression has been linked to increased perioperative pain severity and opioid use (8–10). Catastrophizing is also associated with greater acute postoperative pain, while heightened emotional reactivity to bodily

sensations predicts pain trajectories and the development of widespread pain (11, 12). Despite this extensive literature, most studies examine psychosocial risk factors in isolation or focus primarily on pain intensity, leaving open questions about how these factors relate to widespread pain.

In clinical populations, widespread pain is associated with poorer psychological functioning, delayed recovery, increased opioid use, and higher disability (4, 13). Individuals whose pain extends beyond the surgical site often report feeling confused, disbelieved, or dismissed, complicating both diagnosis and treatment (14, 15). These clinical challenges reflect a deeper theoretical question concerning how pain is encoded, maintained, and modulated within the nervous system (16). Central sensitization—a heightened responsiveness of central nociceptive pathways—has emerged as one construct that may help explain variability in pain persistence and amplification, including postoperative pain intensity and interference (11). However, widespread pain likely reflects the interaction of multiple neurobiological, psychological, and systemic factors rather than any single mechanism (16–18).

Consistent with this view, a growing body of evidence suggests that widespread pain is associated with a combination of neurobiological factors (such as central sensitization), psychological factors (including depression, stress, and catastrophizing), and systemic influences (such as inflammation and metabolic load) (14, 17). Genetic and epigenetic susceptibility, sleep disturbance, and mood symptoms may further interact in shaping the progression from acute to chronic pain and from localized to widespread pain (19). Importantly, much of this evidence comes from community-based or mixed pain populations, with relatively few focused investigations in surgical cohorts, where the timing and location of tissue injury are more clearly defined (5).

Recent population-based work has begun to address these gaps. Using data from the UK Biobank, Tanguay-Sabourin and colleagues identified six modifiable factors associated with widespread pain: body mass index (BMI), fatigue, depression, neuroticism, sleep disturbance, and life stress (20). Together, these factors form a putative Risk of Pain Spread (ROPS) framework, representing one of the first large-scale efforts to quantify psychosocial and behavioral correlates of widespread pain in the general population. Whether this framework generalizes to clinical populations facing an imminent surgical procedure, however, remains unknown.

In the present study, we examined whether psychosocial and sociodemographic factors associated with the degree of widespread pain in population-based samples generalize to clinical populations facing imminent surgical injury. Using two pre-surgical cohorts drawn from the Acute to Chronic Pain Signatures (A2CPS) Consortium (21, 22), we asked how these factors relate to pain intensity at surgical and non-surgical sites and to widespread chronic pain across distinct surgical contexts. We hypothesized that core ROPS factors would show consistent associations with pain intensity, whereas associations with the degree of widespread pain would exhibit greater cohort sensitivity. This approach allows us to evaluate the robustness, generalizability, and clinical relevance of population-derived risk models in pre-surgical populations.

RESULTS

Participant characteristics

Participant characteristics are summarized in Table 1 and Supplementary Figure S1. Females comprised 55% of the Thoracic surgery cohort and 62% of the TKA cohort, reflecting a significantly different female-to-male ratio between cohorts ($\chi^2(1) = 5.78$, $p = 0.02$). TKA participants were older (mean $\pm$ SD: 65.20 ± 8.50 years vs. 59.80 ± 13 years; $t(449.45) = 7.22$, $p = 2.27e-12$), more educated (modal educational attainment: Postgrad vs. High School; $\chi^2(6) = 53.19$,

$p = 1.07e-09$), more likely to be employed ($\chi^2(3) = 8.49$, $p = 0.04$), more likely to be in a higher income bracket (modal income bracket: \$100k-\$149.9k vs. \$50k-\$74.9k; $\chi^2(10) = 50.69$, $p = 2.00e-07$), and less likely to have applied for disability ($\chi^2(2) = 8.42$, $p = 0.02$). No significant differences were observed in marital status between cohorts ($\chi^2(6) = 1.79$, $p = 0.94$).

Among ROPS factors, thoracic surgery participants scored higher on neuroticism ($t(510.87) = 3.80$, $p < .001$, $g = 0.25$) and depression ($t(507.4) = 3.27$, $p < .001$, $g = 0.21$) than TKA participants. Conversely, TKA participants had higher BMI (31.25 vs. 29.33 ; $t(470.58) = 4.83$, $p < .001$, $g = 0.32$), greater fatigue ($t(521.88) = 2.04$, $p = .04$, $g = .13$), and marginally more sleep problems ($t(539.4) = -1.73$, $p = .08$) than Thoracic surgery participants.

Using the Brief Pain Inventory-Short Form (BPI-SF) to assess worst pain intensity over the past 24 hours, TKA participants reported higher [pre-]surgical-site pain (hereinafter SSP - mean $\pm$ SD: 6.16 ± 2.59 vs. 0.99 ± 2.18 ; $t(679.12) = 35.67$, $p < 8.39e-16$) and greater nonsurgical-site pain (NSSP - 4.10 ± 2.93 vs. 2.18 ± 2.82 ; $t(562.71) = 10.45$, $p < 1.82e-23$) compared with Thoracic surgery participants. Widespread chronic widespread pain (Figure 3B) was defined as the number of body regions (out of nine) with pain intensity ≥ 3 sustained for ≥ 6 months using the Michigan Bodymap. Individuals were categorized into one of five levels (0, 1, 2, 3, or ≥ 4 body sites in chronic pain). Individuals with TKA were more likely to report pain in one and two-to-three body sites ($\chi^2(4) = 96.69$, $p = 2.22e-16$), whereas the prevalence of four or more sites (i.e., widespread pain) did not differ significantly.

Latent variable analysis associating ROPS with pain outcomes

To test whether a latent Risk of Pain Spread (ROPS) construct predicts pain burden, we fit a multivariate structural equation model with a single latent ROPS factor predicting log-SSP, log-NSSP, and widespread pain, while adjusting for age, income, sex, marital status, and education. The ROPS latent factor was specified by the six ROPS indicators: adverse childhood experiences, body mass index, neuroticism, sleep quality, fatigue, and depressive symptoms.

Measurement model

All ROPS indicators loaded significantly on the ROPS latent construct (β : 0.18–0.78; $ps < .001$). Loadings were strongest for fatigue ($\beta = 0.74$), depressive symptoms ($\beta = 0.73$), and neuroticism ($\beta = 0.68$), with more modest contributions from sleep quality ($\beta = 0.58$), adverse childhood experiences ($\beta = 0.42$), and BMI ($\beta = 0.18$). Model-based reliability of the latent factor was acceptable ($\omega \approx 0.61$), indicating that the ROP construct captured shared variance across heterogeneous risk indicators.

Structural associations with pain outcomes

In the full multivariate model, the latent ROPS factor significantly predicted all three pain outcomes after adjusting for demographic covariates. Higher ROPs scores were associated with greater SSP ($\beta = 0.22$, $p < .001$), NSSP ($\beta = 0.29$, $p < .001$), and widespread pain ($\beta = 0.27$, $p < .001$). The model explained 9.0% of the variance in pain severity, 11.2% in pain interference, and 9.3% in widespread pain. Residual correlations among pain outcomes were moderate to large (all $ps < .001$), supporting the use of a multivariate outcome model.

Cohort-specific effects

To evaluate whether the ROPS–pain associations differed by clinical cohort, we conducted a multi-group SEM comparing Thoracic surgery and TKA cohorts. Measurement invariance testing

supported equivalence of factor loadings across cohorts (metric invariance: $\Delta\chi^2(5) = 5.20, p = .39$), indicating that the ROPS construct was psychometrically comparable. Constraining all structural paths to equality across cohorts did not significantly worsen model fit ($\Delta\chi^2(24) = 29.91, p = .19$), suggesting no global violation of structural invariance. Despite this, cohort-specific estimates revealed systematically stronger ROPS–pain associations in the Thoracic surgery cohort. In Thoracic participants, the latent ROPS factor showed large, standardized associations with SSP ($\beta = 0.34$), NSSP ($\beta = 0.37$), and widespread pain ($\beta = 0.39$), whereas corresponding effects in the TKA cohort were more modest (β s = 0.20–0.24 for severity and interference; $\beta = 0.21$ for widespread pain). The latent ROPS factor explained a greater proportion of variance in all outcomes in the Thoracic surgery cohort (R^2 range: 13–20%) than in the TKA cohort (R^2 range: 6–11%).

Tests of cohort differences and moderation

Pairwise comparisons of cohort-specific path coefficients indicated that differences between cohorts were marginally different for widespread pain ($\Delta b = 0.45, z = 1.91, p = .056$), with smaller and non-significant differences for SSP and NSSP ($ps > .14$). To formally test moderation, regression models using latent ROPS factor scores revealed significant ROPS $\times$ cohort interactions, demonstrating attenuation of ROPS effects in the TKA cohort. Interaction effects were observed for SSP ($\beta = -0.38, p = .044$) and were strongest for widespread pain ($\beta = -0.77, p < .001$), with a marginally significant effect for NSSP ($\beta = -0.32, p = .081$).

Variable correlogram, dendrogram, and tanglegram analysis

A bivariate Spearman’s correlogram was computed separately for each cohort, revealing higher NSSP for the TKA cohort among females, higher SSP for both cohorts and higher NSSP for the TKA cohort among younger individuals, higher NSSP and spread for unmarried or unpartnered individuals in the TKA cohort, and higher pain outcomes for those with lower income in the TKA cohort. Depression, neuroticism, sleep problems, and fatigue formed a robust ROPS cluster that positively correlated with all pain outcomes for the TKA cohort. For the Thoracic surgery cohort, neuroticism did not significantly correlate with SSP, and depression, neuroticism, and sleep did not significantly correlate with NSSP, but all of them correlated with spread. ACEs and BMI positively correlated with NSSP and spread for the TKA cohort but not the Thoracic surgery cohort. Fatigue was most consistently positively correlated with all pain outcomes for both cohorts. Among both cohorts, the correlations between SSP-NSSP and NSSP-CPS were significantly positively correlated, but SSP was only significantly correlated with spread in the Thoracic surgery cohort.

Tanglegram analysis of the bivariate correlations showed that both cohorts showed highly similar clustering structures (Figure 3 $\epsilon = 0.02$; Baker’s $\gamma = 0.52, p = .0038$). ROPS variables, other than BMI, and pain outcomes formed stable clusters in both the Thoracic and TKA cohort but resided in the same cluster for the Thoracic surgery cohort and bifurcated for the TKA cohort. The sociodemographic variables and BMI grouped together as they shared a residual space of low correlations with the ROPS and pain variables.

Main effects analysis of ‘single-variable regression’ models showed that on average, greater SSP was apparent for those of younger age (-8% , $95\%CI = [-12\%, -3\%]$, $p < .001, R^2 = 0.589$), lower income (-7% , $[-13\%, -0.06\%]$, $p = .048, R^2 = 0.586$), higher ACEs ($+6\%$, $[0.40\%, +13\%]$, $p = .036, R^2 = 0.579$), depression ($+21\%$, $95\%CI = [+15\%, +28\%]$, $p < .001, R^2 = 0.599$), Fatigue ($+25\%$, $[+18\%, +32\%]$, $p < .001, R^2 = 0.611$), Neuroticism ($+14\%$, $[+7\%, +21\%]$, $p < .001, R^2 = 0.579$), and Sleep problems ($+21\%$, $[+14\%, +28\%]$, $p < .001, R^2 = 0.594$). Greater NSSP

and spread was additionally apparent for female sex (NSSP: +36%, [+15%, +62%], $p < .001$, $R^2 = 0.133$; spread: +19%, [+5%, +36%], $p = .008$, $R^2 = 0.011$). Those with lower than an undergraduate education also exhibited greater NSSP (Undergraduate vs. Less than High School: -22%, [-35%, -6%], $p = .008$, $R^2 = 0.128$; Graduate vs. Less than High School: -29%, [-42%, -13%], $p < .001$, $R^2 = 0.128$). Those who were married or domestically partnered exhibited less widespread pain than those who are not married (-33%, [-46%, -16%], $p < .001$, $R^2 = 0.016$).

TKA participants reported substantially higher pain than Thoracic participants: +346% ([301%, 397%], $p < .001$) more SSP +87% ([61%, 117%], $p < .001$) more NSSP. Among the participants who reported chronic pain, the number of chronic pain sites between TKA and Thoracic participants were not significantly different ($p = .069$). For the Thoracic surgery cohort, greater SSP was predicted by lower age (-8% [-12%, -3%], $p < .001$), and lower income (-7% [-13%, -0.06%], $p = .048$). Among ROPS variables, increases in ACEs (+6% [0.40%, 13%], $p = .036$), neuroticism (+14% [+7%, +21%], $p < .001$), sleep problems (+21% [14%, 28%], $p < .001$), fatigue (+25% [18%, 32%], $p < .001$), and depression (+21% [+15%, +28%], $p < .001$) were associated with greater SSP. In multiple regression, only greater fatigue remained associated with higher SSP in the Thoracic surgery cohort (+33.3% [+14.5%, +55.1%], $p < .001$, VI = 100).

Greater NSSP for the Thoracic surgery cohort was predicted by female sex (+36% [12%, 66%], $p < .001$) and lower income (-20% [-28%, -12%], $p < .001$). Like surgical site pain, among ROPS variables, increases in ACEs (+14% [5%, 23%], $p = .002$), depression (+27% [+18%, +37%], $p < .001$), fatigue (+37% [27%, 48%], $p < .001$), neuroticism (+19% [10%, 29%], $p < .001$), and sleep problems (+28% [18%, 39%], $p < .001$) were associated with greater NSSP. Increases in BMI were marginally associated with greater NSSP (+9% [-0.53%, 19%], $p = .065$). In multiple regression, lower income (-15.6% [-25.7%, -4.1%], $p = .010$, VI = 77.71) and greater fatigue (+39.9% [+16.0%, +68.6%], $p < .001$, VI = 100) remained associated with higher NSSP in the Thoracic surgery cohort. Finally, higher spread for the Thoracic surgery cohort was predicted marginally by female sex (+26% [-0.08%, 60%], $p = .051$), lower income (17% [4%, 29%], $p = .013$), and unmarried/unpartnered (vs. married/partnered: 22% [0.76%, 38%], $p = .042$). Among ROPS variables, increases in depression (+24% [+14%, +34%], $p < .001$), fatigue (+19% [+8%, +31%], $p < .001$), neuroticism (+25% [+13%, +37%], $p < .001$), and sleep problems (+20% [+9%, +33%], $p < .001$) was associated with higher spread. Furthermore, increases in ACEs were marginally associated with higher spread (+8% [-0.01%, +17%], $p = .050$). In multiple regression, only greater fatigue (+28.6% [+4.5%, +58.4%], $p = .018$, VI = 100) remained associated with more chronic widespread pain in the Thoracic surgery cohort, although lower income exhibited a marginal association (-19.0% [-35.5%, +1.7%], $p = .070$, VI = 73.64).

For the TKA cohort, greater SSP was also predicted by female sex (+11% [+2%, +21%], $p = .009$), lower age (-8% [-12%, -5%], $p < .001$), lower income (-7% [-11%, -4%], $p < .001$), single individuals (vs. married/partnered: Married/Domestic Partner: -8% [-16%, -0.29%], $p = .041$), less than high school education (vs. undergraduate education: -19% [-34%, -2%], $p = .027$; vs. graduate education: -23% [-37%, -6%], $p = .006$), and a significantly bigger impact than for the Thoracic surgery cohort: vs. undergraduate x cohort -24% [-42%, -2%], $p = .038$). Among ROPS variables, increases in ACEs (+4% [0.32%, 8%], $p = .033$), depression (+10% [+6%, +14%], $p < .001$), fatigue (+15% [11%, 20%], $p < .001$), neuroticism (+7% [3%, 11%], $p < .001$), and sleep problems (+9% [5%, 13%], $p < .001$) were associated with greater surgical site pain. The depression (-10% [-15%, -4%], $p = .002$), fatigue (-8% [-14%, -1%], $p = .017$), and sleep problems (-9% [-15%, -3%], $p = .004$) associations were significantly lower for the TKA cohort relative to the Thoracic surgery cohort. Unique to the TKA cohort, increases in BMI were also associated with greater SSP (+6% [2%, 10%], $p = .001$). In multiple regression, lower age (-6.4% [-10.0%, -2.7%], $p < .001$, VI = 76.02), lower income (-4.9% [-9.2%, -0.3%], $p = .036$, VI

= 46.78), and greater fatigue (+13.2% [+6.8%, +20.0%], $p < .001$, VI = 100) remained associated with greater SSP in the TKA cohort.

Greater NSSP for the TKA cohort was predicted by female sex (+31% [17%, 47%], $p < .001$), lower age (-9% [-14%, -4%], $p = .001$), lower income (-17% [-21%, -12%], $p < .001$), single individuals (vs. married/partnered: -19% [-28%, -9%], $p < .001$), and less than high school education (vs. undergraduate education: -25% [-43%, -1%], $p = .036$; vs. graduate education: -30% [-47%, -8%], $p = .006$). Like SSP, among ROPS variables, increases in ACEs (+9% [3%, 14%], $p = .002$), BMI (+12% [6%, 18%], $p < .001$), depression (+22% [+16%, +28%], $p < .001$), fatigue (+30% [23%, 37%], $p < .001$), Neuroticism (+14% [9%, 20%], $p < .001$), and sleep problems (+15% [9%, 21%], $p < .001$) were associated with greater NSSP. The sleep problems association was significantly lower for the TKA cohort relative to the Thoracic surgery cohort (-10% [-18%, -1%], $p = .025$). In multiple regression, female sex (+16.1% [+2.3%, +31.8%], $p = .017$, VI = 67.28), lower age (-6.8% [-11.7%, -1.7%], $p = .010$, VI = 62.77), lower income (-12.0% [-17.5%, -6.1%], $p < .001$, VI = 97.12), and greater fatigue (+18.0% [+8.7%, +28.0%], $p < .001$, VI = 100) remained associated with greater NSSP in the TKA cohort.

Finally, greater spread for the TKA cohort was predicted marginally by female sex (+13% [-0.73%, 28%], $p = .068$), lower income (-12% [-16%, -7%], $p < .001$), unmarried/unpartnered (vs. married/partnered: -16% [-25%, -6%], $p = .001$), and less than high school education (vs. high school education: -22% [-41%, 3%], $p = .094$; vs. undergraduate education: -32% [-47%, -12%], $p < .001$; vs. graduate education: -33% [-48%, -13%], $p < .001$), which was significantly greater than the Thoracic surgery cohort (vs. high school education x cohort: -43% [-63%, -12%], $p = .011$; vs. undergraduate education x cohort: -49% [-67%, -22%], $p = .002$).

Among ROPS variables, increases in ACEs (+9% [+3%, +15%], $p = .001$), depression (+13% [+8%, +19%], $p < .001$), fatigue (+17% [+11%, +23%], $p < .001$), and sleep problems (+9% [+3%, +14%], $p = .001$) was associated with more widespread chronic pain. Greater neuroticism was also associated with higher spread (+11% [+6%, +17%], $p < .001$), but to a lesser degree than the Thoracic surgery cohort (-10% [-20%, -0%], $p = .043$). Unlike the Thoracic surgery cohort, greater BMI was also associated with higher chronic widespread pain (+6% [0.88%, +12%], $p = .022$). In multiple regression, lower income (-9.1% [-15.3%, -2.4%], $p = .008$, VI = 100), and marginally, greater fatigue (+8.4% [-0.1%, +17.5%], $p = .052$, VI = 84.42) remained associated with greater chronic widespread pain in the TKA cohort.

Network analysis

To examine directional dependencies among sociodemographic, ROPS, and pain variables, we estimated Bayesian network (BN) structures separately for the Thoracic surgery and TKA cohorts as directed-acyclic graphs (DAGs, Figure 6).

In the Thoracic surgery DAG, fatigue emerged as a central hub (Figure 7A), receiving direct input from income, depression, and sleep quality, and in turn influencing SSP, NSSP, and spread. Fatigue and neuroticism were close to pain outcomes (~0.18–0.55), while demographic predictors (sex, age, education, marital) were more distal. SSP and NSSP carried relatively high eigenvector centrality (structurally important sinks). Fatigue and Sleep Quality dominated authority scores. Neuroticism also served as a key connector. Sleep Quality mediated links from ACEs and Neuroticism to Depression and Fatigue, while Neuroticism mediated links from ACEs to BMI, Depression, Sleep Quality, and widespread pain. Pain outcomes clustered together, with directed connections from SSP, to NSSP, and lastly to spread.

In contrast, the TKA DAG showed a more tightly integrated network. Fatigue again occupied a central position, dominating all metrics (pagerank, hub score, eigenvector; Figure 7B) and directly predicting SSP and NSSP, and spread. Fatigue was predicted by ACEs, BMI, and neuroticism,

and in turn predicted depression and sleep quality. NSSP was predicted by age and income as well, which in turn predicted SSP and spread. Income and marital status were surprisingly proximal as well (~ 0.17 – 0.58). This suggests that socioeconomic variables are more entangled in the TKA rather than the Thoracic surgery cohort.

Directionally-constrained model comparison

We compared four directionally-constrained models that differed in how relations between fatigue and pain variables were specified. These models were designed to evaluate whether the observed covariation structure was more consistent with fatigue functioning as an upstream driver, a downstream consequence, or a mutually determined correlate of pain outcomes: (i) the original Fatigue→Pain [F2P] set, in which fatigue is treated as a proximal driver of pain, (ii) Pain→Fatigue [P2F] in which pain severity and pain widespread are assumed to influence fatigue and (iii) unconstrained [UNC] models, in which fatigue–pain relations are determined by structure learning without directional restrictions. We also tested (iv) a fatigue-as-outcome [FOUT] specification to evaluate whether the observed centrality of fatigue could be explained solely by its role as a downstream consequence rather than a driver within the network. Model comparison was done using the conditional Gaussian BIC (BIC-cg; higher values indicate better fit).

In the Thoracic surgery cohort, F2P and UNC were essentially tied (BIC-cg = -7065.445 for both; $\Delta\text{BIC} = 0$), suggesting that the data do not meaningfully distinguish whether Fatigue→Pain constraints improve fit over the baseline constraints. P2F fit ($\Delta\text{BIC} = -0.040$) and FOUT fit ($\Delta\text{BIC} = -1.70$) was worse. In the TKA cohort, F2P also provided the best fit (BIC-cg = -22652.485), with UNC ($\Delta\text{BIC} = -0.61$), P2F ($\Delta\text{BIC} = -5.24$), and FOUT ($\Delta\text{BIC} = -20.80$) being worse. These findings indicate that modeling pain as a downstream outcome of fatigue—rather than modeling fatigue as downstream of pain or as the ultimate outcome—is more consistent with the observed covariation structure in both clinical cohorts.

Taken together, these networks highlight fatigue as a consistent central determinant across cohorts, bridging sociodemographic and psychosocial predictors with pain outcomes. Neuroticism and depression further reinforced these pathways, particularly in the TKA cohort where BMI also contributed. The cross-cohort comparison suggests that while fatigue universally links upstream vulnerability factors to pain, the embedding of BMI and mood symptoms into the pain network may be stronger in individuals undergoing TKA.

Mediation models

Mediation models informed by the F2P DAGS were tested to test indirect effects on pain outcomes, and the results are summarized in Table 2. Average Indirect Effects (AIE), Average Direct Effects (ADE), and Proportion Mediated (PM) were estimated using nonparametric bootstrapping. For interpretability, pathways were categorized as: full mediation (AIE significant; ADE not significant), partial (complementary) mediation (AIE and ADE significant and same sign), competitive (suppression) (AIE and ADE significant with opposite signs), direct-only (ADE significant; AIE not), total-only (TE significant; AIE and ADE not), or no effect (none significant). Because multiple hypothesis-driven pathways were prespecified by DAGs, we emphasize effect sizes and interval estimates; no multiplicity correction was applied.

Thoracic surgery cohort

Across pain outcomes for the thoracic surgery cohort, fatigue emerged as the dominant statistical mediator, particularly for depression and sleep problems. For SSP and NSSP, the associations of both depression and sleep problems with pain were largely or fully explained by indirect effects through fatigue, with negligible direct effects once fatigue was included (Table 2). Household income showed limited or inconsistent indirect effects through fatigue for SSP and NSSP, and in several cases operated primarily through direct or alternative pathways. For chronic widespread

pain, both NSSP and fatigue served as key intermediaries. The effect of SSP on widespread pain was fully mediated by NSSP, whereas fatigue exerted both direct and indirect effects on spread through NSSP. Using fatigue as the mediator, depression showed a strong indirect association with widespread pain, whereas household income exhibited primarily direct effects. When neuroticism was modeled as the mediator, age and ACEs influenced widespread pain predominantly through neuroticism, while household income showed partial mediation. Taken together, the Thoracic cohort results indicate a multistep structure in which psychosocial vulnerability factors influence fatigue and internalizing ROPS traits, which in turn shape pain intensity and ultimately widespread pain.

TKA cohort

In the TKA cohort, mediation patterns were more centralized around fatigue and NSSP. For SSP, the effects of age and fatigue were partially mediated by NSSP, whereas income effects were largely indirect. For both SSP and NSSP, income, neuroticism, and ACEs showed substantial indirect effects through fatigue, with weak or non-significant direct effects (Table 2). For NSSP, socioeconomic indicators, including marital status and educational attainment, were primarily associated with pain via income, which functioned as a key intermediary. For chronic widespread pain, NSSP fully mediated the effects of income and age, while fatigue showed both direct and indirect associations with spread. When fatigue was specified as the mediator, income, adverse childhood events, and neuroticism were predominantly linked to widespread pain through indirect pathways. Overall, the TKA cohort exhibited a more tightly integrated mediation structure, with fatigue and non-surgical pain acting as central conduits through which sociodemographic, and ROPS factors related to pain outcomes.

DISCUSSION

Include a Discussion that summarizes your conclusions and elaborates on their implications. There should be a paragraph outlining the limitations of your results and interpretation, as well as a discussion of the steps that need to be taken for the findings to be applied in the clinic. Please avoid claims of priority and avoid repeating the conclusions at the end.

Across two large, well-characterized pre-surgical cohorts from the Acute to Chronic Pain Signatures (A2CPS) Data Release 2.0.0, our findings converge on fatigue as a central organizing feature of pain burden. Fatigue emerged as the strongest and most consistent contributor to pain across outcomes, including surgical-site pain (SSP), non-surgical-site pain (NSSP), and chronic widespread pain, and remained robust after adjustment for sociodemographic, and Risk of Pain Spread (ROPS) psychodemographic covariates. These results extend prior work implicating fatigue as a core feature of centralized and widespread pain (5, 14, 20, 23), and suggest that fatigue may function as a proximal expression of cumulative biopsychosocial risk rather than a secondary consequence of pain alone. Consistent with this interpretation, multiple upstream factors—including the other ROPS factors: sleep disturbance, depressive symptoms, neuroticism, adverse childhood experiences (ACEs), body mass index (BMI) – as well as socioeconomic disadvantage—were linked to pain outcomes largely through their shared variance with fatigue or through NSSP en route to widespread pain. At a higher level of abstraction, these relationships were captured by a latent Risk of Pain Spread (ROPS) construct that robustly predicted SSP, NSSP, and widespread pain across cohorts. Importantly, while the ROPS construct was psychometrically stable across clinical groups, its association with pain outcomes was substantially stronger in the Thoracic surgery cohort than in the TKA cohort, particularly for widespread pain. The weaker associations observed in TKA participants may reflect the presence of strong nociceptive drivers that partially mask the influence of underlying vulnerability factors, whereas the associations exhibited by Thoracic surgical participants may more directly reflect centrally mediated processes. Overall, these findings support a model in which fatigue is a central

component of pain vulnerability, while underscoring meaningful differences in how latent risk manifests across pain etiologies.

The pattern across regression, mediation, and network analyses supported a two-stage pathway from upstream risk to widespread pain. First, psychosocial load (socioeconomic stress, ACEs, neuroticism) potentially elevates fatigue (4, 19). Second, fatigue more likely increases pain intensity (SSP and NSSP) than the other way around, with NSSP serving as a proximal conduit to widespread pain in both cohorts (16, 17). This aligns with broader models that view widespread pain as the product of distributed dysregulation, not localized nociception alone, and helps explain why reducing site-specific nociceptive drive is often insufficient to prevent spread (3).

Network structures highlighted cohort-specific topologies. In Thoracic participants, the network was more distributed, with fatigue, sleep, neuroticism, and depression acting as multiple, moderately central mediators. In TKA participants, the system was more centralized and “fatigue-centric”, with fatigue dominating graph metrics and income appearing unusually proximal to pain outcomes. Concomitantly, higher BMI, lowered sleep quality, and increased depression appear to manifest in parallel with pain symptoms as a result of a combination of low income and fatigue.

Mediation results largely tracked estimated network edges, increasing confidence in the directional hypotheses encoded by the DAGs (24). In Thoracic participants, sleep and depression effects on SSP/NSSP were fully mediated by fatigue, and $SSP \rightarrow NSSP \rightarrow$ spread captured a plausible route from local to widespread pain. In TKA, income, neuroticism, and ACEs influenced pain through fatigue, and marital/education effects on NSSP operated via household income. Importantly, NSSP often fully or partially mediated effects on spread, reinforcing the role of generalized pain intensity as the step immediately preceding spread (2).

Lower income is consistently related to higher NSSP and greater spread—directly in Thoracic and via fatigue or NSSP in TKA—implicating socioeconomic stressors in both intensity and spread (15). Younger age predicted higher pain in both cohorts (more strongly in TKA), and female sex was associated with higher NSSP (in both cohorts) and marginally higher spread. Together, these gradients argue that Risk of Pain Spread is not solely biological; it is shaped by socioeconomic position and role demands that likely interact with sleep, mood, and energy regulation (1, 20).

Three clinical practice takeaways emerge:

1. Make fatigue a focus of treatment in individuals with chronic pain. Given its centrality and consistent mediation, routine fatigue assessment and fatigue-directed interventions (e.g., sleep regularization, graded activity pacing, treatment of mood or sleep disorders, metabolic optimization) may yield downstream benefits on pain intensity and function, and ultimately contain spread (13, 14, 23, 25).
2. Monitor and treat NSSP as a harbinger of spread. In both cohorts, NSSP mediated effects on spread. Systematically measuring and managing non-index-site pain during recovery may interrupt escalation from localized to widespread pain (16).
3. Communicate and address present or potential socioeconomic headwinds. Income-linked effects—direct or routed via fatigue/NSSP—suggest that embedding social risk screening and support (benefits navigation, return-to-work planning, financial counseling) into perioperative care might attenuate modifiable drivers of spread (5).

Methodological notes

Strengths of this series of analyses include (i) two large pre-surgical cohorts with harmonized pain phenotyping (BPI-SF, Michigan Body Map), (ii) theory-guided DAGs that constrained Bayesian network learning and mediation, and (iii) convergent evidence across complementary models. Limitations include (i) observational design and reliance on sequential ignorability for

causal mediation; (ii) potential measurement error in self-reports (sleep, fatigue, mood); (iii) listwise deletion for missingness; and (iv) scale dependencies for PM under log links.

Our analytic approach conceptually diverges from the conclusions of (20). Unlike their work, which empirically derived single-variable dichotomous (yes/no) predictors of ROPS, we employed continuous, scale-based proxies of the predictors of interest, allowing us to capture well-validated gradations in risk rather than categorical thresholds. Our results also diverged in that the ROPS predictors we examined did not equally or simultaneously impact spread; when they were related to spread at all, their effects were indirect, operating primarily through fatigue and/or NSSP. Moreover, rather than focusing on contemporaneous stress exposures as in Tanguay-Sabourin, we operationalized early adversity through ACEs, which capture developmental risk factors occurring much earlier in the lifespan. This shifts the interpretive frame from stress as an ongoing proximal trigger of widespread pain toward adversity as a distal vulnerability factor that may shape later pain susceptibility. A further limitation of our work is that we do not currently include a measure of contemporaneous life stress, which constrains direct comparison with their findings. Regression effects were reported as percent change to preserve interpretability and consistency across latent-variable, single-variable, multiple-regression, network, and mediation analyses strengthens inference. Nevertheless, non-normality, heteroscedasticity, and influential observations persisted for some outcomes, so we therefore emphasize effect sizes and CIs over binary significance and interpret suppression/competitive mediation patterns cautiously (26). Importantly, these comparisons do not establish causal direction but instead evaluate which directional assumptions are most compatible with the observed multivariate dependency structure under plausible clinical constraints. Causal interpretation relies on standard sequential ignorability assumptions: no unmeasured confounding of predictor–mediator, predictor–outcome, and mediator–outcome relations (conditional on measured covariates), no mediator–outcome confounders affected by the predictor, correct model specification, consistency, and no interference. Future work ideally will incorporate pre-/post-operative longitudinal sampling, objective sleep/actigraphy to anchor objective sleep metrics with fatigue, inflammatory and metabolic biomarkers, and interventional tests that manipulate fatigue or sleep to evaluate causal leverage on widespread pain (3, 17).

In two pre-surgical cohorts, we identified a fatigue-centered, intensity-to-spread pathway that integrates psychosocial and sociodemographic risk with clinical pain outcomes. While details differ by pre-surgical context, the recurring motif is that upstream psychosocial vulnerabilities heighten fatigue, fatigue amplifies pain intensity (especially beyond the target site), and increasing NSSP presages widespread pain. This framework yields concrete, testable targets—fatigue and NSSP—for preventing the transition from localized to widespread pain during surgical recovery (20).

MATERIALS AND METHODS

All descriptions of Materials and Methods should be included in the main paper. The Materials and Methods should be broken up into sections, each with a short subheading. Please include a study design paragraph at the start of the Materials and Methods and a statistical paragraph at the end of the Materials and Methods in the main text. If the Materials and Methods make the paper exceed the length limitations, less important sections of the Materials and Methods can be moved to the Supplementary Materials.

Experimental Design

We analyzed baseline data from two independent pre-surgical cohorts drawn from the A2CPS Consortium – Data Release 2.0.0, which included participants preparing for total knee arthroplasty (TKA) and participants preparing for thoracic surgery. These cohorts differ in surgical indication, pain burden, and demographic composition, allowing us to examine whether

associations between psychosocial risk factors and pain outcomes generalize across distinct pre-surgical contexts. We focused on three pain outcomes assessed prior to surgery: worst intensity of surgical-site pain (SSP), worst intensity of non-surgical-site pain (NSSP), and chronic widespread pain, defined as the number of body regions with moderate or greater pain persisting for at least six months. Predictors included sociodemographic variables (age, sex, marital status, income, and education) and psychosocial and behavioral factors corresponding to the Risk of Pain Spread (ROPS) framework, including body mass index, fatigue, depression, neuroticism, sleep disturbance, and adverse childhood experiences. We examined associations between predictors and pain outcomes using complementary analytic approaches. We first estimated latent-variable and regression-based models to characterize univariate and multivariable associations. We then used network and mediation analyses to assess patterns of statistical dependence among variables and to identify factors occupying central positions within association networks. We conducted all analyses separately within each cohort and compared results across cohorts to assess robustness and cohort sensitivity.

Baseline A2CPS assessment and sample inclusion criteria

Data collection involved harmonized, standardized protocols across study sites, including self-report questionnaires, clinical evaluations, and behavioral measures. The shared and cohort-specific eligibility criteria can be found in Table 1 of Berardi, et al., 2022 (21). Participants were not excluded for prior chronic pain conditions, including fibromyalgia, neuropathic pain, peripheral neuropathy, low back pain, mental health conditions, or prior surgery > 3 months previously; or based on prior or baseline pain medication use, including opioids, to maximize generalizability. A full Schedule of Activities can be found in Supplementary Table S3 of Berardi et al., 2022 (21). Inclusion criteria for the present analysis were complete baseline phenotyping data and availability of key sociodemographic and ROPS variables. Participants were excluded if they withdrew consent or were deemed ineligible by site investigators.

With the sparse sample size across multiple levels of each demographic variable, we dropped the Intersex/Unknown (n = 15) levels of the sex variable, dropped the Unknown education level (n = 32), aggregated the marital status variable to Non-Married, Married/Domestic Partner, and dropped the Unknown marital status (n = 31) level, and converted the income status variable into a continuous one by setting each participant's income status to the median of their reported income range.

Each participant in the Thoracic surgery and TKA cohorts was categorized into pain subtypes using the BPI and MBM (Figure 2, Methods). Individuals were classified as either having No Chronic Pain or Chronic Pain (any pain duration ≥ 3 months as measured by the MBM). Within the No Chronic Pain category, subtypes included: no pain reported at all (No Pain), no pain in the last 24 hours (as measured by the BPI; No Pain < 24hrs), minimal pain in the last 24 hours (pain intensity ≤ 3 ; Minimal Pain < 24hrs) or some acute pain in the last 24 hours (Acute Pain < 24hrs). Those in the Chronic Pain category were further divided into two subtypes: Low Chronic Pain (pain intensity ≤ 3 and any pain duration ≥ 3 months) and Moderate/High Intensity Chronic Pain (pain intensity > 3 and pain duration ≥ 3 months in at least one body site).

Pain intensity and spread – Brief Pain Inventory and Michigan Body Map

Pain was assessed using the Brief Pain Inventory (BPI; 27) and the Michigan Body Map (MBM; 28). The BPI was modified to assess the worst pain intensity at the pre-surgical site over the past 24 hours with anchors of 0 (no pain) to 10 (worst pain imaginable). The Michigan Body Map was used to record the number of painful body regions with persistent pain over the past 3 months (at baseline).

ACEs

The Adverse Childhood Experience questionnaire (ACE; 29) is a 10 item retrospective assessment of childhood (age 0–18 years) abuse, neglect, and household dysfunction. Each item is scored dichotomously (0 = absent, 1 = present) and summed to an ACE total score ranging from 0 to 10. Higher scores reflect greater cumulative exposure to early adversity.

Neuroticism

The Neuroticism (Negative Emotionality) subscale was derived from the Big Five Inventory–2 short form (BFI-2-S; 30), comprising of four items: “Is temperamental, gets emotional easily,” and reverse codings of “Is emotionally stable, not easily upset,” “Is relaxed, handles stress well”, and “Is persistent, works until the task is finished.” Participants rated each item according to their level of agreement from 1 (disagree strongly) to 5 (agree strongly). Responses to these items were averaged to yield a mean Neuroticism score, scored out of 20, with higher values indicating greater emotional instability and sensitivity to stress.

Depression

Depression was assessed using the Patient Health Questionnaire-8 (PHQ-8; 31), which included questions about participant experiences during the past 2 weeks, scored from 0 (“not at all”) to 3 (“nearly every day”). Higher total scores represented elevated levels of depression.

Sleep Disturbance

Sleep disturbance is assessed using the Patient Reported Outcomes Measurement Information System (PROMIS) Short Form v1.0 – Sleep Disturbance 6a measure (32). Subjects rate their sleep quality and difficulty falling asleep over the past 14 days on a scale of 1 (not at all) to 5 (very much) with higher scores indicating greater sleep disturbance.

Fatigue

Fatigue was assessed using the PROMIS Short Form v1.0 – Fatigue 7a (33). This seven-item measure has Likert-scale responses ranging from 1 (never) to 5 (always) with higher scores indicating greater fatigue.

Latent variable analyses

We used structural equation modeling (SEM) with lavaan (*sem()*) to define a latent Risk of Pain Spread (ROPS) construct and examine its association with SSP, NSSP, and widespread pain. The latent ROPS factor was specified using the six observed ROPS predictors. All continuous variables were standardized prior to modeling. Models adjusted for demographic covariates, accounted for correlated pain outcomes, and handled missing data using full information maximum likelihood with robust estimation. Each outcome was modeled simultaneously, with residual correlations specified between pain outcomes to account for shared variance not explained by the latent factor. Cohort differences were evaluated using multi-group SEM and nested model comparisons.

Correlogram, dendrogram, and tanglegram analyses

Pairwise correlations among predictors and outcomes were computed using Spearman’s coefficients and visualized with correlograms. Hierarchical clustering was applied to correlation matrices, and dendrograms were compared across methods using tanglegrams to evaluate cluster stability. Correlograms were generated separately for each cohort and variables were clustered using Ward’s minimum variance method (Ward.D2). To compare the resulting dendrograms across cohorts, we used tanglegrams, where colors indicated cluster membership. We computed two complementary metrics: entanglement, which quantifies how well two trees can be aligned without crossing edges (0 = identical, 1 = maximally tangled), and Baker’s γ correlation, which

assesses concordance of ultrametric distances between the trees (−1 to 1). Significance of Baker's γ was evaluated using 5,000 permutations of the dendrogram labels to generate a null distribution.

Single-variable and multivariable regression analyses

Associations between predictors and pain phenotypes were examined using generalized linear models. Ordinary Least Squares (OLS) multiple regression models were first fit within the Thoracic surgery and TKA participant cohorts to assess model assumptions. Residuals and leverage values were extracted from each model to evaluate multicollinearity (Variance Inflation Factor - VIF), heteroskedasticity (Breusch–Pagan χ^2 - BP χ^2), non-normality (Shapiro–Wilk W - SW-W), influence (Cook's D), and a global omnibus test of skewness, kurtosis, and functional form (Global Validation of Linear Model Assumptions; using the *gvlma* package in R).

Regression diagnostics indicated distinct patterns of model assumption violations across cohorts and outcomes. In both cohorts, VIFs showed no multicollinearity issues. The SSP model showed the most severe assumption breaches, with heteroscedasticity (BP $\chi^2 = 43.66$, $p < .001$), non-normality (SW-W $p < .001$), and violations of skewness and kurtosis ($ps < .05$), alongside several influential observations. For NSSP, both heteroscedasticity (BP $\chi^2 = 31.58$, $p = .005$) and non-normal residuals (SW-W $p < .001$) were present, with multiple influential cases; global tests again suggested skewness departures. For spread, residual variance was stable (BP $\chi^2 = 17.41$, $p = .18$), though residuals deviated from normality (SW-W $p = .003$), and several observations were highly influential. Global validation tests further flagged skewness violations. In the TKA cohort, the SSP model showed stable variance (BP $\chi^2 = 12.66$, $p = .70$), but residuals were non-normal (SW-W $p < .001$) and influenced by a small number of high-leverage points. By contrast, for NSSP, residuals were heteroscedastic (BP $\chi^2 = 30.84$, $p = .014$) as well as heavy-tailed (SW-W $p = .004$), with kurtosis flagged in global tests. For widespread pain, residuals were also heteroscedastic (BP $\chi^2 = 36.20$, $p = .003$), non-normal (SW-W $p < .001$), and flagged for skewness, kurtosis, and link function misspecification ($ps < .05$).

Given consistent OLS violations across cohorts involving non-normal residuals, influential cases, and global test failures related to skewness and heavy tails, OLS regression models were performed on log-transformed ($\log(1+y)$) outcomes on the worst pain ratings for both SSP and NSSP. Coefficients were back-transformed so that for each 1-unit increase in a predictor we report the percent change in worst pain: $\Delta\% = (e\beta - 1) \times 100\%$. Quasipoisson regression (log link) was used for the count of spread given its negative skew. We likewise report percent change using $(e\beta - 1) \times 100\%$, which is equivalent to $(\text{Incidence Rate Ratio IRR} - 1) \times 100\%$. McFadden R^2 was computed for quasipoisson models as a proxy for variance explaining spread. First, “single-predictor” regression models were fit where each outcome was regressed upon by a single predictor, cohort (effects coded centered at 0 with Thoracic set as -1, and TKA as 1), and the predictor x cohort interaction. This was followed by multiple regression models where all predictors were fit simultaneously in an independent model for each cohort. Multiple regression models were ten-fold cross-validated to estimate the Variable Importance (VI) out of 100 of each predictor's inclusion in each model. Employment status and application for disability were not tested as predictors because of their likelihood of codetermination with pain symptom outcomes. Predictor variables were explicitly coded as either discrete (categorical) or continuous (numeric) to permit estimation under a conditional Gaussian framework. Categorical demographic variables (e.g., sex, marital status, education) were treated as factors, while continuous measures (e.g., age, income, ACEs, BMI, neuroticism, sleep quality, fatigue, depression, spread and log-transformed SSP and NSSP) were modeled as numeric.

Network analyses

To assess conditional dependencies between predictors and outcomes, we estimated partial correlation networks using the graphical least absolute shrinkage and selection operator

(GLASSO) with extended Bayesian information criterion (EBIC) model selection to improve model sparsity. Network centrality indices (strength, closeness, betweenness) were computed to identify influential variables. Cases with missing data were excluded listwise. We applied a score-based tabu search algorithm (bnlearn v.4.9; 34) using the Bayesian Information Criterion for conditional Gaussian networks (BIC-cg) as the model selection score. To enforce valid conditional Gaussian models, edges from continuous to discrete variables were blacklisted, preventing Gaussian parents of categorical children. Additional blacklisted constraints were incorporated where theoretically justified e.g., demographic variables set as exogenous, pain outcome variables spread, SSP, and NSSP were set as endogenous, and endogenous edges to household income, age, BMI, ACEs, and neuroticism from sleep quality, fatigue, and depression were blacklisted. To interpret the networks, computed centrality metrics included indegree, outdegree, total degree, betweenness centrality, closeness, authority, pagerank centrality, eigenvector centrality, hubscore, and distance to outcome.

Directionally-constrained model comparison

To evaluate competing directional hypotheses between fatigue and pain, we constructed four model classes: Fatigue→Pain (F2P), Pain→Fatigue (P2F), Unconstrained (UNC), and Fatigue-as-Outcome (FOUT) by restricting or permitted specific edges using whitelists (required directions) and blacklists (forbidden directions). All models enforced plausible constraints preventing reverse causation into trait-like variables (e.g., age, sex, income) and disallowing continuous→categorical edges to satisfy the conditional Gaussian (CG) model assumptions. Each model class was fit separately within the Thoracic surgery and TKA cohorts. To assess relative evidence for each directional hypothesis, BIC-cg scores were compared between models, where higher values indicate better model fit after penalizing complexity. We summarized model fit across classes by computing ΔBIC relative to the best model for each cohort and ranked models accordingly. To evaluate the robustness of key directional edges, we repeated structure learning using nonparametric bootstrapping ($R = 5,000$) with the same whitelist/blacklist constraints and computed arc strength and direction probabilities using *boot.strength*.

Mediation analyses

Mediation models were estimated using the mediation and lavaan R packages. Average Indirect Effects (AIE), Average Direct Effects (ADE), and Proportion Mediated (PM) were quantified, with 5,000 bootstrap samples used to generate bias-corrected 95% confidence intervals. Mediation pathways tested whether ROPS predictors influenced spread indirectly via fatigue or NSSP.

For continuous outcomes (SSP and NSSP), we fit observed-variable path models using SEM with lavaan (*sem()*), specifying two regression equations per mediation:

1. $M \sim X + \text{covariates}$ (paths “a”)
2. $Y \sim M$ (path “b”) + X (paths “c”) + covariates.

We used quasipoisson models for spread and a linear model for the mediator and effects are reported on the expected count scale. For DAG indicated pathways within each cohort, we estimated:

1. natural indirect effects ($\text{AIE} = E[Y(x, M(x)) - Y(x, M(x'))]$)
2. direct effects ($\text{ADE} = E[Y(x, M(x')) - Y(x', M(x'))]$),
3. total effects (Total Effect: $\text{TE} = \text{AIE} + \text{ADE}$), and
4. Proportion mediated (at control n_0): $\text{PM} = \text{AIE} / \text{TE}$.

95% confidence intervals (CIs) were obtained via nonparametric bootstrap with bias-corrected and accelerated (BCa) intervals. We targeted 5,000 resamples. Mediation effects (AIE, ADE, TE) were estimated with the mediation package (*mediate()*).

Statistical Analysis

We analyzed baseline data from two independent pre-surgical cohorts (thoracic surgery $n = 358$; TKA $n = 1043$) and conducted all analyses separately within each cohort. Sample sizes (n), statistical tests, and exact P values for individual analyses are reported in the corresponding figure legends and main text. All statistical analyses were conducted in R (version 4.5.1) using packages including lavaan, igraph, qgraph, and corplot. We used two-sided statistical tests throughout defining statistical significance using an alpha level of 0.05. Effect sizes (e.g., β coefficients, incident rate ratios, partial R^2) and 95% confidence intervals are reported alongside p-values to facilitate interpretation when needed. We assessed bivariate associations using Spearman rank correlations to accommodate non-normal distributions. To evaluate associations between sociodemographic and psychosocial predictors and pain outcomes, we fit regression models appropriate to each outcome type, including linear and generalized linear models. We standardized continuous predictors to facilitate comparison of effect sizes across variables. Multivariable models included sociodemographic covariates and psychosocial predictors entered simultaneously to estimate independent associations. We used latent-variable modeling to estimate shared variance among related psychosocial measures and to reduce measurement noise. We applied network-based analyses to characterize patterns of statistical association among variables and to identify factors with high centrality within association networks. We conducted mediation analyses to quantify the extent to which associations between predictors and widespread pain were statistically accounted for by intermediate pain measures. These analyses represent statistical decomposition of associations and do not imply causal relationships.

Ethical approval

All procedures were approved by a single Institutional Review Boards (IRBs) at the University of Iowa. All participants provided written informed consent in accordance with the Declaration of Helsinki.

Data and materials availability

All data are available in the main text or the supplementary materials. All data provided from the A2CPS consortium are available to other investigators online upon permission granted by <https://a2cps.org/researchers/accessing-our-data/>. Detailed code and annotation will be made publicly available at GitHub (<https://github.com/canlab> and <https://github.com/a2cps>).

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Funding:

National Institutes of Health administrative supplement as part of grant 5U54DA049110-04 (FOA RFA-RM-18-031) (MS, MAL)

National Institutes of Health Common Fund, Office of the Director/Office of Strategic Coordination (A2CPS Consortium)

National Institutes of Health grant U24NS112873 (Clinical Coordinating Center)

National Institutes of Health grant U54DA049110 (Data Integration and Resource Center)

National Institutes of Health grant U54DA049116 (Omics Data Generation Center)

National Institutes of Health grant U54DA049115 (Omics Data Generation Center)

National Institutes of Health grant U54DA049113 (Omics Data Generation Center)

National Institutes of Health grant UM1NS112874 (Multi-site Clinical Center 1)

National Institutes of Health grant UM1NS118922 (Multi-site Clinical Center 2)

Author contributions:

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Investigation: MS, TDW

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Writing—review & editing: MS, TDW, BA, DC, REH, KS, CMK, CMB, MAL

Competing interests: Chad Brummett serves as a consultant for Vertex Pharmaceuticals and Merck Pharmaceuticals and provides expert medicolegal testimony. Tor Wager serves on the scientific advisory board of Curable Health. All other authors declare no competing interests.

Data and materials availability: All data are available in the main text or the supplementary materials. All data provided from the A2CPS consortium are available to other investigators online upon permission granted by <https://a2cps.org/researchers/accessing->

our-data/. Detailed code and annotation will be made publicly available at GitHub (<https://github.com/canlab> and <https://github.com/a2cps>).

Figures

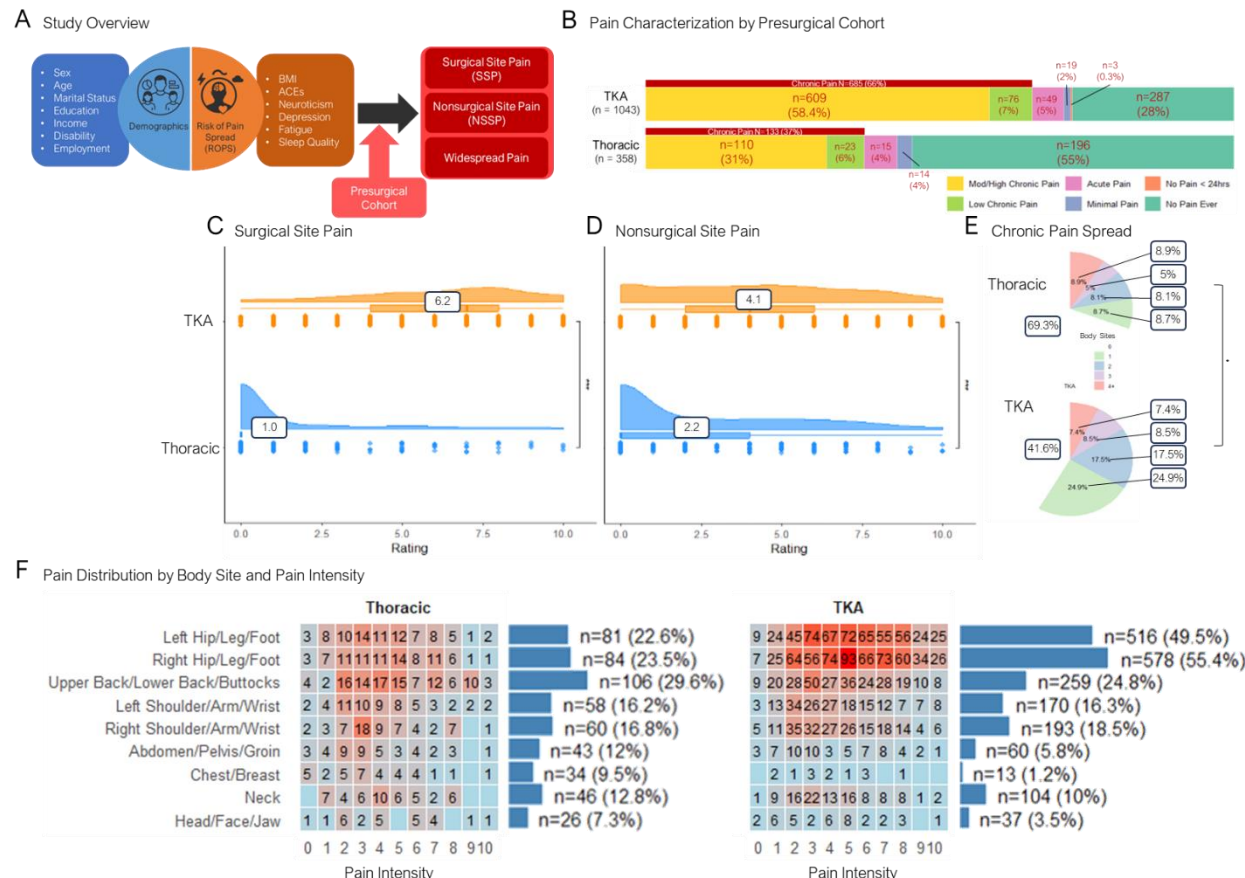


Fig. 1. Study design and baseline pain characteristics across surgical cohorts. (A) Study overview and analytical framework. Presurgical pain outcomes—surgical-site pain intensity (SSP), non-surgical-site pain intensity (NSSP), and widespread pain—were modeled as a function of sociodemographic variables and analogues of Risk of Pain Spread (ROPS) predictors, with cohort (thoracic surgery vs. total knee arthroplasty [TKA]) examined as a potential moderator. **(B) Distribution of pain subtypes across cohorts using the Michigan Body Map.** Pain ratings ≥ 3 defined moderate-to-high pain. Pain was classified as chronic if reported to last ≥ 3 months. **(C–E) Cohort distributions of presurgical pain outcomes.** Shown are distributions of (C) SSP, (D) NSSP, and (E) widespread pain. TKA participants reported higher SSP and NSSP and a greater number of chronic pain sites among individuals with chronic pain. TKA participants were more likely than thoracic surgery participants to report one to three painful sites ($\chi^2(4) = 96.69$, $p = 2.22 \times 10^{-16}$), whereas rates of four or more painful sites (widespread pain) did not differ significantly between cohorts. **(F) Widespread pain by body site and intensity.** Heatmap shows the number of participants reporting pain at each body site and intensity level (any duration), stratified by cohort.

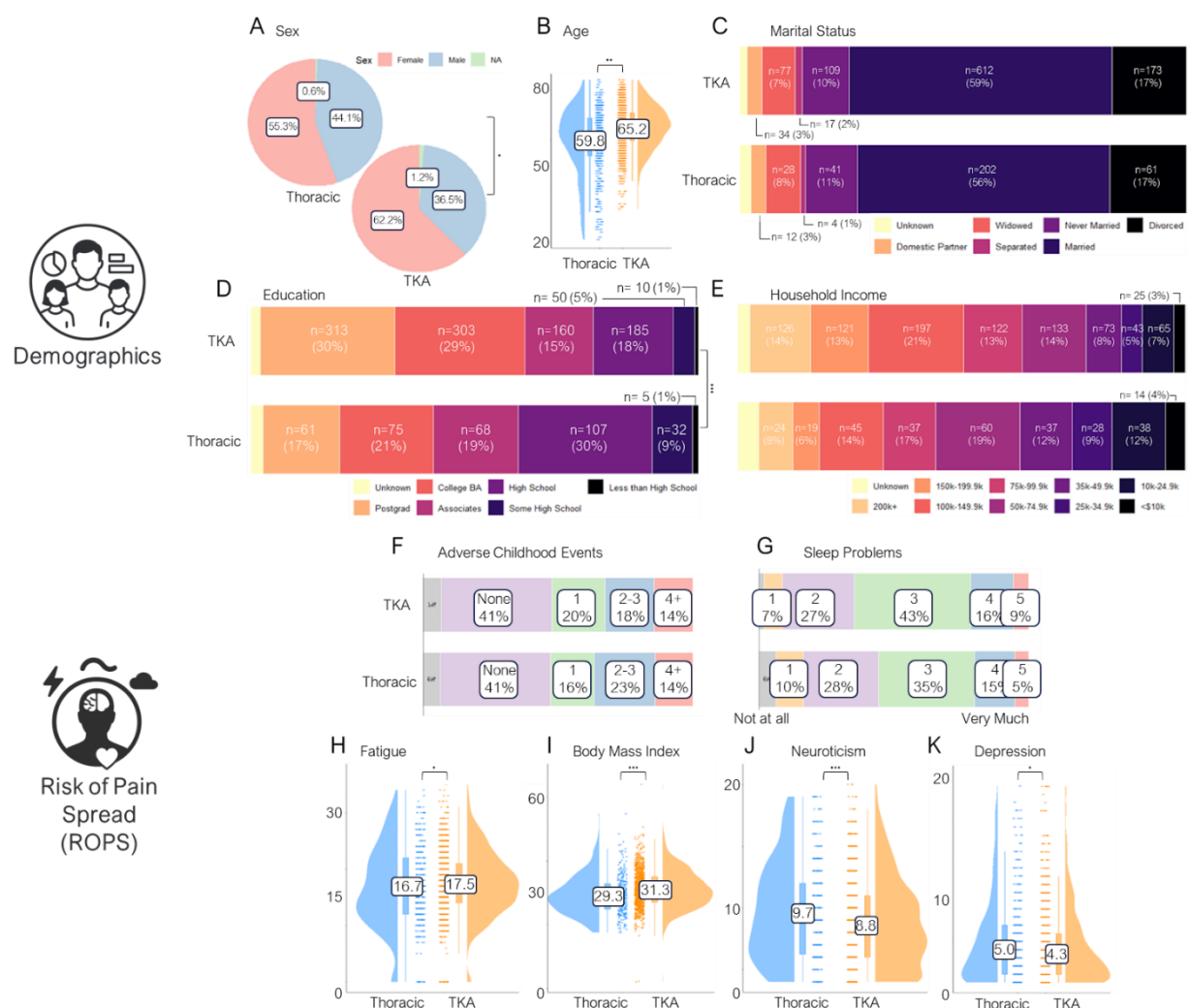


Fig. 2. Sociodemographic and psychosocial differences across surgical cohorts. (A–E) Sociodemographic characteristics by cohort. The thoracic surgery and total knee arthroplasty (TKA) cohorts differed significantly in sex composition, age, education, and household income. Relative to the thoracic surgery cohort, the TKA cohort was older, more highly educated, and reported higher income. **(F–K) Risk of Pain Spread (ROPS)–related factors by cohort.** The TKA cohort showed higher body mass index and greater fatigue, whereas the thoracic surgery cohort reported higher levels of neuroticism and depressive symptoms.

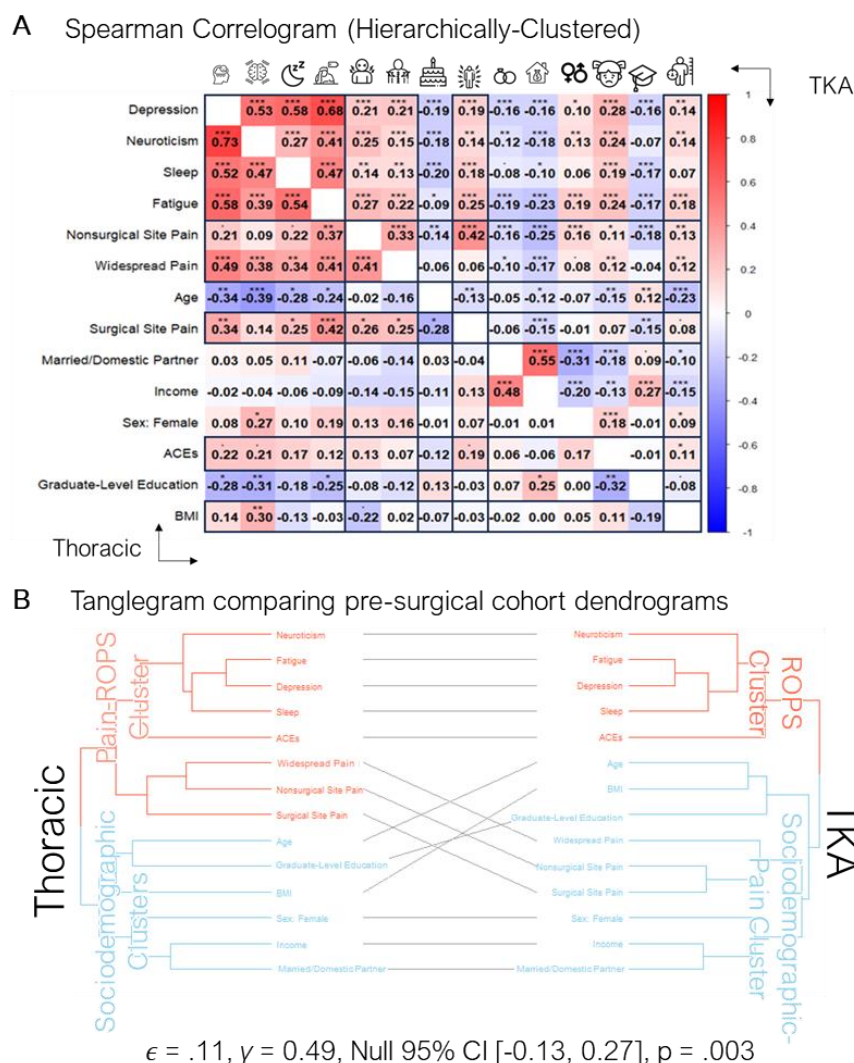


Fig. 3. Similar correlational structure of pain and ROPS variables across cohorts. (A) Spearman correlation structure by cohort. Spearman correlogram showing the thoracic surgery cohort (lower-left triangle) and the total knee arthroplasty (TKA) cohort (upper-right triangle). Variables were hierarchically clustered based on the average correlation structure across cohorts, revealing close overall proximity between Risk of Pain Spread (ROPS) variables and pain outcomes. **(B) Comparison of cohort-specific clustering structure.** A tanglegram contrasts the dendrograms derived separately for each cohort. Low entanglement (0.11) indicates high structural similarity between clustering solutions, and Baker's gamma (0.49; $p = 0.003$) confirms that this similarity is unlikely to arise by chance. In the thoracic surgery cohort, pain outcomes clustered more closely with ROPS variables (excluding BMI), forming a unified cluster distinct from sociodemographic factors. In contrast, in the TKA cohort, ROPS variables (excluding BMI) clustered separately from pain outcomes.

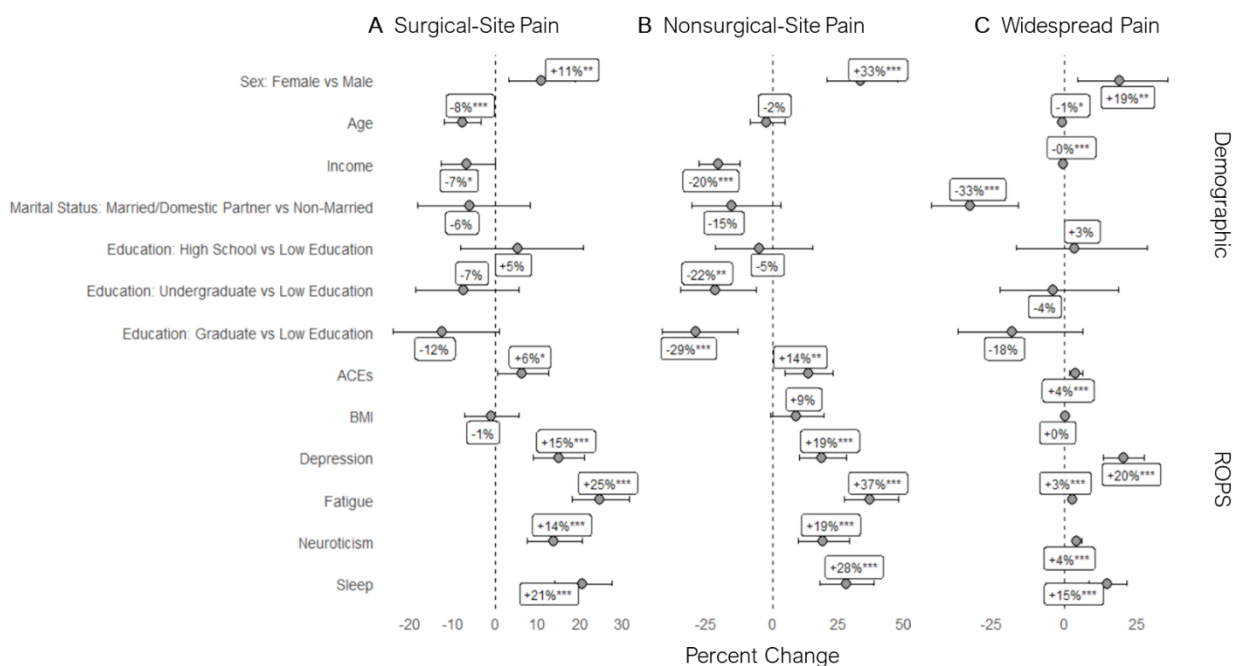
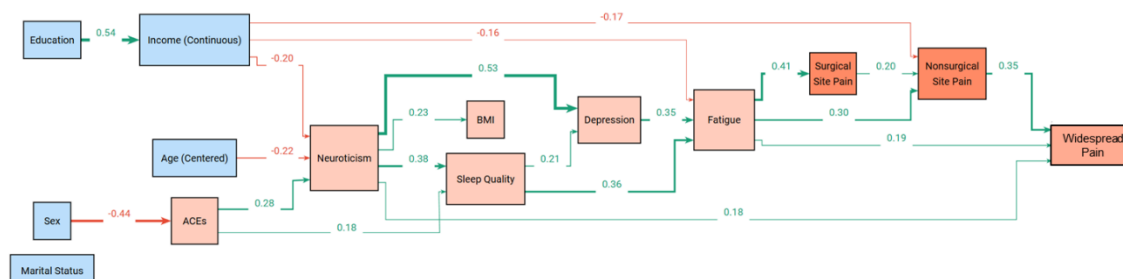


Fig. 4. Single-variable associations of ROPS predictors with presurgical pain outcomes. Standardized effects from single-variable regression models predicting presurgical pain outcomes. Shown are standardized regression coefficients for Risk of Pain Spread (ROPS) predictors and sociodemographic variables predicting (A) surgical-site pain (SSP), (B) non-surgical-site pain (NSSP), and (C) widespread pain. Across outcomes, greater pain severity was associated with higher fatigue, depressive symptoms, adverse childhood experiences (ACEs), neuroticism, and sleep disturbance, as well as lower household income and younger age. Female sex showed stronger associations with NSSP and widespread pain, and lower educational attainment was selectively associated with higher NSSP.



Fig. 5. Univariate and multivariable predictors of presurgical pain outcomes by cohort. Comparison of single-variable and multivariable regression models predicting presurgical pain outcomes. Top panels show standardized effects from single-variable regression models estimated in the combined sample, with cohort-specific means overlaid. Bottom panels show cohort-specific multivariable regression models including all predictors simultaneously. Models predict (A, D) surgical-site pain (SSP), (B, E) non-surgical-site pain (NSSP), and (C, F) widespread pain estimated separately for the thoracic surgery and total knee arthroplasty (TKA) cohorts.

A Thoracic Surgery Cohort Directed Acyclic Graph



B TKA Cohort Directed Acyclic Graph

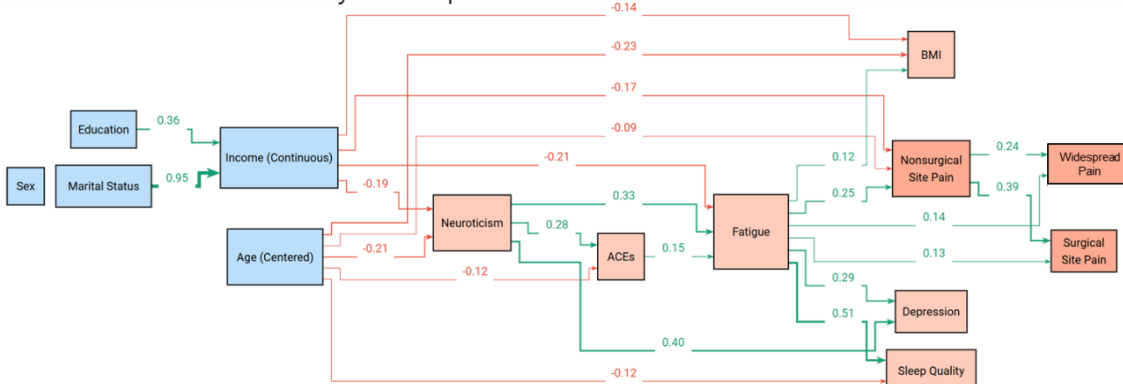
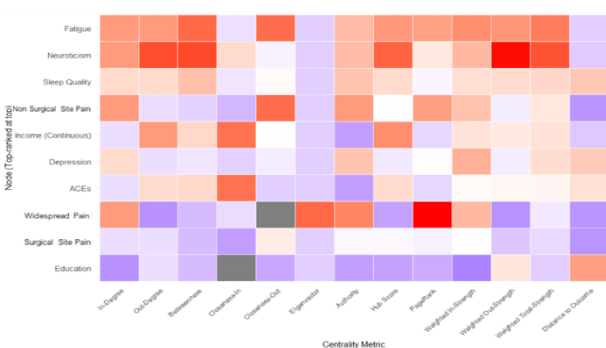


Fig. 6. Fatigue-centered causal network structure of presurgical pain across cohorts. Directed acyclic graphs (DAGs) representing cohort-specific Bayesian network structures. Shown are DAGs estimated separately for the (A) thoracic surgery cohort and (B) total knee arthroplasty (TKA) cohort. Nodes represent sociodemographic (blue), Risk of Pain Spread (ROPS; light orange), and pain outcome (dark orange) variables; arrows indicate conditional dependencies after accounting for all other variables in the network. Across cohorts, fatigue emerged as a central hub linking upstream sociodemographic and psychosocial factors to pain outcomes. Pain outcomes—surgical-site pain (SSP), non-surgical-site pain (NSSP), and widespread pain—formed a tightly connected subnetwork, with cohort-specific differences in connectivity. The TKA network showed greater overall integration, including stronger involvement of BMI, neuroticism, and depression.

A Thoracic surgery cohort Centrality Heatmap



B TKA cohort Centrality Heatmap

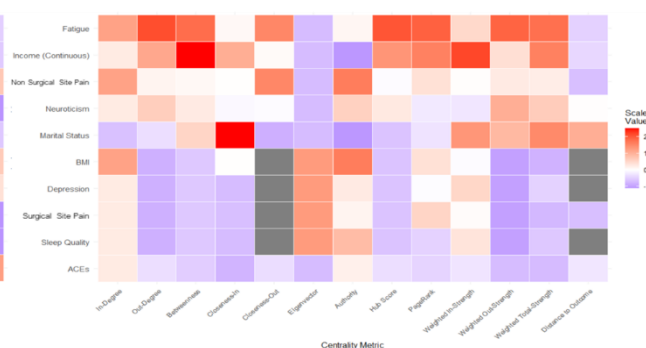


Fig. 7. Fatigue is the most central node in cohort-specific pain networks. Centrality metrics for cohort-specific directed acyclic graphs. Heatmaps display the top 10 most central variables in the Bayesian network models for the (A) thoracic surgery cohort and (B) total knee arthroplasty (TKA) cohort. Centrality measures quantify each variable's relative importance within the network structure. Across both cohorts, fatigue ranked as the most central variable, indicating a prominent role in linking sociodemographic and psychosocial risk factors to pain outcomes.

Table 1. Participant characteristics and baseline pain profiles by surgical cohort. Baseline sociodemographic characteristics, Risk of Pain Spread (ROPS) factors, and pain measures for total knee arthroplasty (TKA) and thoracic surgery cohorts. Continuous variables are reported as mean (SD) unless otherwise noted; categorical variables are reported as percentages or modal category. Group differences were assessed using Welch's t tests for continuous variables and χ^2 tests for categorical variables. Pain duration is reported as median and mean (SD). widespread pain reflects the number of body sites with pain intensity ≥ 3 lasting ≥ 3 months.

	TKA	Thoracic	Test Statistic	p-value
N	1043	358		
Demographics				
Age in years Mean (SD)	65.2 (8.5)	59.8 (13)	$t(449.45) = 7.22$	$2.27e-12^{***}$
Sex (Mode)	Female	Female	$\chi^2(1) = 5.78$	0.02^*
% Male	36.5	44.1		
% Female	62.2	55.3		
Marital Status (Mode)	Married	Married	$\chi^2(6) = 1.79$	0.94
Income Level (Mode)	100k-149.9k	50k-74.9k	$\chi^2(10) = 50.69$	$2e-7^{***}$
Education Level (Mode)	Postgrad	High School	$\chi^2(6) = 53.19$	$1.07e-9^{***}$
Employment Status (Mode)	Not Employed	Not Employed	$\chi^2(3) = 8.49$	0.04^*
Applied for Disability (Mode)	No	No		
% Applied for Disability	11.1	15.9	$\chi^2(2) = 8.42$	0.02^*
Risk of Pain Spread (ROPS)				
Adverse Childhood Events (ACEs)	1.51	1.67	$t(537.65) = 1.15$	0.25
Body Mass Index (BMI), kg/m ²	31.25	29.33	$t(470.58) = 4.83$	$1.88e-6^{***}$
Depression (PHQ score)	4.34	5.01	$t(523.93) = 2.16$	.03*
Fatigue (PROMIS score)	17.53	16.68	$t(521.88) = -2.04$	.04*
Neuroticism (BIG5 score)	8.80	9.73	$t(510.87) = 3.80$	$1.62e-4^{***}$
Sleep Problems (PROMIS score)	2.87	2.76	$t(539.4) = -1.73$	.08 [†]
Pain				
Pain Duration Median	36 months	1 month	$t(654.90) = 6.02$	$2.93e-9^{***}$
Mean (SD)	71.4 (173.8)	17.1 (120.4)		
Surgical Site Pain Intensity (SSP)	6.16	0.99	$t(679.12) = -35.67$	$8.39e-158^{***}$
Nonsurgical Site Pain Intensity (NSSP)	4.10	2.18	$t(679.12) = -35.67$	$1.82e-23^{***}$
Widespread Pain	1.51	1.11	$t(679.12) = -3.65$	$2.85e-4^{***}$
Widespread Pain (Modal Body Sites)	0	0	$\chi^2(6) = 7.29$	0.03^*
Widespread Pain (%0 Body Sites)	41.6	69.3		
Widespread Pain (%1 Body Sites)	24.9	8.7		
Widespread Pain (%2 Body Sites)	17.5	8.1		
Widespread Pain (%3 Body Sites)	8.5	5		
Widespread Pain (%4+ Body Sites)	7.4	8.9		

[†] $p < .10$, * $p < 0.05$, ** $p < 0.01$ and *** $p < .001$.

Table 2. Directed acyclic graph–guided mediation analyses of presurgical pain outcomes. Shown are mediation models linking predictors (X) to pain outcomes (Y) through DAG-consistent mediators (M), estimated separately for thoracic surgery and total knee arthroplasty (TKA) cohorts. Indirect effects are reported as average indirect effects (AIE), direct effects as average direct effects (ADE), and total effects with 95% confidence intervals. Proportion mediated reflects the fraction of the total effect attributable to the indirect pathway. Mediation type indicates full, partial, or direct-only mediation based on the significance of indirect and direct paths. All models adjusted for cohort-specific covariate structure implied by the DAGs.

Cohort	Predictor (X)	Mediator (M)	Outcome (Y)	Indirect		Direct		Total		Proportion mediated [§]	Mediation type
				AIE (95% CI)	p	ADE (95% CI)	p	Total (95% CI)	p		
Thoracic	Depression	Fatigue	Widespread Pain	0.30 [0.15, 0.53]	0.002**	0.11 [-0.07, 0.30]	0.232	0.46 [0.26, 0.71]	< 0.001***	63.5%	Full Mediation
Thoracic	Income	Fatigue	Widespread Pain	-0.00 [-0.01, 0.00]	0.187	-0.02 [-0.04, -0.00]	0.052 [†]	-0.02 [-0.05, -0.01]	0.045*	17.1%	Direct-only Effect
Thoracic	Sleep problems	Fatigue	Widespread Pain	0.07 [0.02, 0.14]	0.031*	0.05 [0.00, 0.08]	0.021*	0.13 [0.10, 0.16]	< 0.001***	46.8%	Partial Mediation
Thoracic	Fatigue	Non-surgical-site pain intensity	Widespread Pain	0.01 [0.00, 0.01]	< 0.001***	0.01 [0.00, 0.01]	0.002**	0.02 [0.01, 0.02]	< 0.001***	40.8%	Partial Mediation
Thoracic	Income	Non-surgical-site pain intensity	Widespread Pain	-0.00 [-0.01, -0.00]	< 0.001***	-0.01 [-0.03, -0.00]	< 0.001***	-0.02 [-0.04, -0.00]	< 0.001***	28.8%	Partial Mediation
Thoracic	Surgical-site pain intensity	Non-surgical-site pain intensity	Widespread Pain	0.09 [0.05, 0.14]	< 0.001***	0.06 [-0.01, 0.13]	0.135	0.15 [0.08, 0.23]	< 0.001***	54.3%	Full Mediation
Thoracic	Age (centered)	Neuroticism	Widespread Pain	-0.01 [-0.01, -0.00]	0.011*	0.01 [-0.01, 0.02]	0.541	-0.00 [-0.02, 0.01]	0.815	56.6%	Full Mediation
Thoracic	Income	Neuroticism	Widespread Pain	-0.00 [-0.01, -0.00]	0.078 [†]	-0.01 [-0.03, -0.00]	0.076 [†]	-0.02 [-0.04, -0.00]	0.056 [†]	15.5%	Partial Mediation
Thoracic	Adverse Childhood Events	Neuroticism	Widespread Pain	0.04 [0.02, 0.08]	0.010**	0.04 [-0.04, 0.10]	0.303	0.08 [0.01, 0.15]	0.022*	47.7%	Full Mediation
Thoracic	Depression	Fatigue	Surgical-site pain intensity	0.08 [0.03, 0.15]	0.007**	0.06 [-0.07, 0.20]	0.374	0.14 [0.02, 0.27]	0.023*	57.4%	Full Mediation
Thoracic	Income	Fatigue	Surgical-site pain intensity	-0.00 [-0.00, 0.00]	0.152	-0.00 [-0.00, 0.00]	0.522	-0.00 [-0.00, 0.00]	0.298	39.7%	No effect
Thoracic	Sleep problems	Fatigue	Surgical-site pain intensity	0.06 [0.02, 0.11]	0.010*	0.06 [-0.03, 0.15]	0.201	0.12 [0.04, 0.21]	0.005**	50.8%	Full Mediation
Thoracic	Depression	Fatigue	Non-surgical-site pain intensity	0.11 [0.05, 0.18]	0.002**	0.00 [-0.15, 0.16]	0.955	0.11 [-0.04, 0.25]	0.131	96%	Full Mediation
Thoracic	Income	Fatigue	Non-surgical-site pain intensity	-0.00 [-0.00, 0.00]	0.210	-0.00 [-0.00, -0.00]	0.004**	-0.00 [-0.01, -0.00]	0.002**	11.4%	Direct-only Effect
Thoracic	Sleep problems	Fatigue	Non-surgical-site pain intensity	0.09 [0.04, 0.15]	0.002**	0.11 [-0.01, 0.22]	0.062 [†]	0.19 [0.09, 0.30]	< 0.001***	44.2%	Full Mediation
Thoracic	Fatigue	Surgical-site pain intensity	Non-surgical-site pain intensity	0.01 [0.00, 0.02]	0.017*	0.04 [0.03, 0.06]	< 0.001***	0.05 [0.04, 0.07]	< 0.001***	18.1%	Partial Mediation
TKA	Income	Fatigue	Widespread Pain	-0.00 [-0.00, -0.00]	0.001**	-0.00 [-0.00, 0.00]	0.972	-0.00 [-0.00, 0.00]	0.428	56.9%	Full Mediation
TKA	Adverse Childhood Events	Fatigue	Widespread Pain	0.03 [0.01, 0.04]	< 0.001***	-0.00 [-0.05, 0.04]	0.907	0.02 [-0.02, 0.07]	0.312	75.4%	Full Mediation
TKA	Neuroticism	Fatigue	Widespread Pain	0.01 [0.01, 0.02]	< 0.001***	0.01 [-0.01, 0.03]	0.270	0.03 [0.01, 0.04]	0.001**	55.6%	Full Mediation
TKA	Age (centered)	Non-surgical-site pain intensity	Widespread Pain	-0.00 [-0.01, -0.00]	0.008**	-0.00 [-0.02, 0.01]	0.352	-0.01 [-0.02, 0.00]	0.106	40.7%	Full Mediation
TKA	Fatigue	Non-surgical-site pain intensity	Widespread Pain	0.01 [0.01, 0.01]	< 0.001***	0.02 [0.01, 0.02]	< 0.001***	0.02 [0.02, 0.03]	< 0.001***	33.3%	Partial Mediation
TKA	Income	Non-surgical-site pain intensity	Widespread Pain	-0.00 [-0.00, -0.00]	< 0.001***	-0.00 [-0.00, 0.00]	0.984	-0.00 [-0.00, 0.00]	0.242	86.4%	Full Mediation
TKA	Income	Fatigue	Surgical-site pain intensity	-0.00 [-0.00, -0.00]	0.005**	-0.00 [-0.00, 0.00]	0.061 [†]	-0.00 [-0.00, -0.00]	0.010*	26.5%	Full Mediation
TKA	Adverse Childhood Events	Fatigue	Surgical-site pain intensity	0.01 [0.00, 0.01]	0.004**	-0.01 [-0.03, 0.01]	0.376	-0.00 [-0.02, 0.02]	0.884	-506.6%	Full Mediation
TKA	Neuroticism	Fatigue	Surgical-site pain intensity	0.01 [0.01, 0.01]	< 0.001***	-0.00 [-0.01, 0.01]	0.645	0.01 [-0.00, 0.02]	0.222	137.5%	Full Mediation
TKA	Age (centered)	Non-surgical-site pain intensity	Surgical-site pain intensity	-0.00 [-0.00, -0.00]	0.009**	-0.00 [-0.01, -0.00]	0.017*	-0.01 [-0.01, -0.00]	< 0.001***	31.9%	Partial Mediation
TKA	Fatigue	Non-surgical-site pain intensity	Surgical-site pain intensity	0.01 [0.01, 0.01]	< 0.001***	0.01 [0.00, 0.02]	0.017*	0.02 [0.01, 0.03]	< 0.001***	47.9%	Partial Mediation
TKA	Income	Non-surgical-site pain intensity	Surgical-site pain intensity	-0.00 [-0.00, -0.00]	< 0.001***	-0.00 [-0.00, 0.00]	0.739	-0.00 [-0.00, 0.00]	0.059 [†]	83.2%	Full Mediation
TKA	Education: Graduate	Income	Non-surgical-site pain intensity	-0.18 [-0.26, -0.10]	< 0.001***	-0.07 [-0.30, 0.16]	0.543	-0.25 [-0.48, -0.02]	0.030*	71%	Full Mediation
TKA	Education: High School	Income	Non-surgical-site pain intensity	-0.03 [-0.07, 0.01]	0.171	-0.02 [-0.25, 0.23]	0.885	-0.04 [-0.28, 0.20]	0.718	60.1%	No effect
TKA	Education: Undergraduate	Income	Non-surgical-site pain intensity	-0.11 [-0.17, -0.06]	< 0.001***	-0.10 [-0.32, 0.13]	0.387	-0.21 [-0.43, 0.01]	0.064 [†]	52.7%	Full Mediation
TKA	Marital Status: Married/Domestic Partner	Income	Non-surgical-site pain intensity	-0.14 [-0.20, -0.08]	< 0.001***	0.03 [-0.09, 0.15]	0.619	-0.11 [-0.22, 0.00]	0.055 [†]	128.3%	Full Mediation

Note: Mediation analyses were restricted to paths supported by cohort-specific DAG structures and should be interpreted as conditional associations rather than definitive causal effects.

† $p < .10$, * $p < 0.05$, ** $p < 0.01$ and *** $p < .001$. §Proportions mediated may exceed 100% or be negative when indirect and direct effects operate in opposite directions.